

Pipe Materials & Types

Copper • CPVC • PEX • PPR • HDPE • Steel • Cast Iron

What You Will Learn

- Eight pipe materials: copper, CPVC, PVC, PEX, PPR, HDPE, galvanised steel, cast iron
- Properties of each material: temperature rating, pressure rating, lifespan, and cost
- Joining methods: solder, push-fit, compression, threaded, and fusion welding
- Material selection guide by application: cold/hot water, drainage, underground, gas

Material Temperature Limits (continuous service)

Copper:	250°C	PPR (PN20):	95°C	CPVC:	93°C
PEX-A:	95°C	uPVC / PVC:	60°C	HDPE:	60°C

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PLUMB-003: Pipe Materials and Types

Choosing the wrong pipe material is one of the most expensive mistakes in plumbing. The wrong plastic in a hot water system degrades and leaks within months. The wrong metal in an aggressive soil causes premature corrosion. The right pipe for the job depends on five factors: the fluid being carried, the temperature and pressure conditions, the installation environment, the joining method available, and the project budget. This lesson covers the eight most widely used plumbing pipe materials and the criteria for selecting between them.

Learning Objectives

- Identify the eight major plumbing pipe materials and their physical properties
- State the maximum temperature and pressure ratings for each material
- Select the correct pipe material for given applications (cold water, hot water, drainage, gas, underground)
- Describe the five main joining methods and which materials each applies to
- Avoid the most common pipe material selection mistakes that cause premature failures

1. Overview of Plumbing Pipe Materials



Material	Colour/Appearance	Max Temp.	Max Pressure	Lifespan	Relative Cost
Copper (Type L)	Bright copper/orange	250°C	15 bar (at 20°C)	50+ years	High
CPVC	Cream/off-white	93°C	15 bar	25–40 years	Medium
uPVC / PVC	White or grey	60°C	16 bar (PN16)	25–50 years	Low

PEX-A (cross-linked)	Red (hot) / Blue (cold)	95°C	10 bar	25–50 years	Medium
PPR (polypropylene)	Green or grey	95°C	20 bar (PN20)	25–50 years	Low-Medium
HDPE / PE100	Black (UV stabilised)	60°C	16 bar (PN16)	50+ years	Medium
Galvanised steel	Silver/grey (zinc coated)	120°C	25 bar	15–30 years	High
Cast iron	Dark grey/black	250°C	16 bar	75–100 years	Very High

2. Copper Pipe — The Traditional Standard

Copper is the oldest and most respected plumbing pipe material. It is naturally antimicrobial (bacteria cannot survive on copper surfaces), corrosion resistant in most environments, and compatible with almost all fittings and joining methods. The main disadvantages are its high cost (copper is a commodity metal with volatile pricing) and the skill required for soldered joints.

Copper Type	Wall Thickness	Pressure Rating	Standard Use
Type K (heavy)	Thickest wall	Highest	Underground, medical gas, high-pressure commercial systems
Type L (medium)	Medium wall	High	Standard domestic and commercial hot/cold water supply — the most common specification
Type M (light)	Thin wall	Medium	Low-pressure domestic supply, HVAC refrigerant (US standard)
Type DWV	Thinnest wall	Low (gravity only)	Drain, waste, and vent (DWV) systems — not for supply pressure

3. Plastic Pipe Materials — PVC, CPVC, PEX, PPR, HDPE

Plastic pipes have replaced copper and steel in most modern residential and commercial installations due to their lower cost, ease of installation, and resistance to corrosion. Each plastic type has a distinct chemical composition that determines its temperature and pressure capability.

Material	Full Name	Key Property	Cannot be Used For
uPVC	Unplasticised Polyvinyl Chloride	Excellent chemical resistance, rigid, cold water and drainage	Hot water supply (softens above 60°C). Cannot be used for hot water.
CPVC	Chlorinated Polyvinyl Chloride	Hot water rated to 93°C. Same ease of installation as PVC.	Not suitable for underground burial without protection. UV degrades over time.
PEX-A	Cross-Linked Polyethylene (Engel method)	Flexible — bends around corners without fittings. Shape memory returns if kinked.	Cannot be joined with solvent cement. Requires special fittings (crimp, clamp, or expansion).
PEX-B	Cross-Linked Polyethylene (silane method)	Cheaper than PEX-A. Slightly less flexible and shape memory.	Some manufacturers do not warrant for potable (drinking) water — check certification.

PPR	Polypropylene Random Copolymer	Hot and cold water. Fusion welded — no chemicals, no solvents.	Not suitable for gas supply. UV degrades — must be insulated or protected outdoors.
HDPE / PE100	High-Density Polyethylene	Underground water mains, sewage, gas. Flexible in large diameters.	Not rated for continuous hot water service above 60°C in pressure applications.

4. Pipe Joining Methods

SOLDERED
(Capillary)

coupling



Copper only.
Heat with torch,
solder wicks in.
Permanent.

PUSH-FIT
(Speedfit)



Any plastic pipe.
Press in — done.
Releasable tool
needed.

COMPRESSION
Fitting

olive



Copper or plastic.
Compress olive
with spanners.
Releasable.

THREADED
(BSP/NPT)



Steel/copper.
Tapered male thread
into female socket.
PTFE tape.

FUSION
Welding

fused



PPR/HDPE/PE.
Heat tool fuses
pipe + fitting.
Permanent.

Method	Materials	Tools Required	Releasable ?	Key Rule
Capillary soldering (hard solder)	Copper only	Blow torch, flux, lead-free solder, pipe cutter, deburring tool	No — permanent	Pipe must be bone dry and clean. Moisture prevents solder from flowing. Use lead-free solder for potable water (95% tin/5% antimony or silver solder).
Push-fit (Speedfit / Hep2O)	Copper, PEX, CPVC (check manufacturer)	Pipe cutter only. Hand pressure to assemble.	Yes — with release tool	Deburr and mark insertion depth. Do not twist after insertion. Check pipe end is square and smooth.
Compression fitting	Copper, PEX with insert, HDPE	Two spanners (one to hold, one to turn). No heat.	Yes — can be reassembled	Do not over-tighten — 1¼ turns after hand-tight is standard for most fittings. Over-tightening deforms the olive and causes leaks.
Threaded (BSP)	Steel, copper, some plastic with inserts	Pipe wrench, thread cutter or stocks and dies, PTFE tape	Yes — but not for frequent disassembly	Wrap PTFE tape clockwise on male thread (3–6 turns). Tighten until snug — over-tightening splits female fittings.
Fusion welding (socket/butt)	PPR, HDPE, PE, PVDF	Electrofusion or socket fusion tool — specific temperature 260°C for PPR	No — permanent molecular bond	Temperature must be exact for the material. Time the heat exposure accurately. Do not twist or move joint until cooling time is complete.

5. Material Selection Guide



6. Steel and Cast Iron Pipes

Material	Types Available	Joining Method	Advantages	Disadvantages
Galvanised steel (GI)	Black steel (uncoated) and hot-dip galvanised. Sizes: ½" to 8" BSP.	Threaded (BSP) with fittings, or welded for larger diameters	High pressure rating. Mechanical strength — resists impact and vibration. Available everywhere.	Corrodes internally over 15–30 years, especially in soft or acidic water. Not suitable for oxygen-rich water systems.
Stainless steel 304	Thin wall (press fit systems: Viega, Geberit) or heavy wall threaded.	Press-fit with special jaw tool, or threaded, or welded (TIG)	Excellent corrosion resistance. Long lifespan. Suitable for potable water, dairy, food industry.	High cost. Press-fit tools are brand-specific and expensive. Thin wall vulnerable to mechanical damage.
Stainless steel 316	As above but with molybdenum addition for marine and chloride environments.	As above	Best corrosion resistance of common grades. Required in coastal, offshore, and chemical applications.	Higher cost than 304. Difficult to distinguish from 304 visually — always verify grade marking.
Cast iron (CI)	Hub-and-spigot (traditional) or plain end with rubber gasket (modern).	Compression rubber gasket joints or lead/oakum caulked (traditional)	Excellent sound dampening — quietest drainage pipe. Very long lifespan (75–100 years). Fire rated.	Very heavy — requires proper support. Brittle — cracks under impact. Specialist labour for installation.

7. Common Pipe Material Mistakes and How to Avoid Them

Mistake	Risk	Correct Practice
Using uPVC for hot water supply	Pipe softens, deforms, and leaks within months above 60°C	Use CPVC, PPR, or copper for all hot water supply lines. uPVC only for cold water and drainage.

Connecting copper directly to galvanised steel	Galvanic corrosion at the bi-metallic junction — steel corrodes rapidly, causing blockage and leaks	Use a dielectric union (insulating fitting) between copper and steel, or use brass fittings as a transition.
Burying standard PVC underground without protection	UV not an issue underground, but soil movement and point loads crack PVC. Wrong class for pressure mains.	Use HDPE or MDPE for underground water mains. If using PVC underground, use Class D (for buried applications).
Using push-fit fittings in inaccessible locations	Push-fit fittings can relax over time — periodic inspection required. Cannot access for re-tightening if buried.	Use soldered copper or fusion-welded PPR for buried or permanently concealed connections.
Applying PVC solvent cement to non-PVC plastic pipe	Solvent cement for PVC does not bond CPVC, ABS, or PPR. Joint appears to hold then fails under pressure.	Use the correct solvent cement specifically formulated for the pipe material. Colour-coded: grey=PVC, orange=CPVC.

8. Practice Exercises

#	Exercise	Scenario / Instructions
1	Material identification	From the following list, identify whether each pipe is suitable for hot water supply (95°C), cold water supply only, drainage only, or underground water mains: (a) uPVC PN10, (b) PPR PN20, (c) Copper Type L, (d) HDPE PE100, (e) Cast iron, (f) CPVC Schedule 80, (g) PEX-A, (h) Galvanised steel BSP.
2	Joining method selection	For each installation situation, state the correct joining method and explain why: (a) Copper 22mm hot water supply in a concealed ceiling void — permanent joint needed. (b) HDPE 63mm underground water main — no open flame allowed on site. (c) 15mm copper at a kitchen under-sink connection where future access is needed. (d) PPR 32mm in a boiler room at 85°C constant temperature.
3	Material for new bathroom	A new 3-bedroom house requires plumbing for: cold water supply throughout, hot water supply from a solar water heater (max 85°C), shower drainage, and underground connection from street main to house (15m run). Specify the pipe material for each section and justify your selection.
4	Fault diagnosis	A plumber installed a hot water system using uPVC PN10 pipe throughout. After 6 months, the pipes are sagging and leaking at joints. Explain: (a) why this failed, (b) what maximum temperature the pipe was rated for, (c) what material should have been used, and (d) what the correct temperature rating should be.
5	Cost vs lifespan calculation	Compare copper Type L and PPR PN20 for a 50-metre domestic hot water run: Copper: £8/m pipe + £3/fitting × 30 fittings = total material cost. PPR: £4/m pipe + £1.50/fitting × 30 fittings + £200 fusion tool hire. Calculate total material cost for each. Copper lifespan: 50 years. PPR lifespan: 25 years. Which is more economical over 50 years (assume PPR replaced once)?

Lesson Summary

Pipe Material	Max Temp	Key Rule
Copper Type L	250°C	Gold standard for durability. Lead-free solder only for potable water. Antimicrobial properties.
uPVC / PVC	60°C	Cold water and drainage ONLY. Never for hot water supply. Use CPVC if you need a plastic for hot water.
CPVC	93°C	Hot water rated plastic. Requires CPVC-specific orange solvent cement — not compatible with PVC cement.
PEX-A	95°C	Flexible — ideal for cramped spaces. Shape memory. Requires crimp, clamp, or expansion fittings — not solvent.
PPR	95°C	Fusion welded — chemical-free joints. Common in MEP and commercial projects. Best where no naked flame allowed.
HDPE/PE100	60°C	Underground water mains and sewage. Electrofusion or butt fusion joining. Highly flexible in large diameters.

Galvanised steel	120°C	Use dielectric union when connecting to copper. Corrodes internally over 15–30 years in soft water.
Cast iron	250°C	Quietest drainage pipe. Extremely heavy. 75–100 year lifespan. Required in some fire-rated applications.

What Is Next — PLUMB-004

PLUMB-004: Pipe Fittings and Joints — The complete reference for plumbing fittings: elbows, tees, couplings, reducers, unions, valves, and adapters. How to specify fittings by material, size, and angle. Reading and drawing plumbing isometric diagrams. Fitting types for each pipe material. Common fitting mistakes and leak points.

Resource	Details
Plumbing PDF Series	PLUMB-001 through PLUMB-030 — ayetechub.com/pdfs.html
MEP Engineering PDFs	Related MEP content — ayetechub.com/pdfs.html
Community	Telegram: t.me/ayetechub